

Power series MethodLegendre's Equation

$$* (1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \text{--- (1)} ; \quad n \in \mathbb{Z}.$$

$$p(x) = \frac{-2x}{1-x^2}$$

$$q(x) = \frac{n(n+1)}{1-x^2}$$

$$\lim_{x \rightarrow 0} p(x) = 0$$

$$\lim_{x \rightarrow 0} q(x) = n(n+1) \Rightarrow x=0 \text{ is an ordinary point.}$$

Let $y = \sum_{m=0}^{\infty} a_m x^m$ be solution of (1).

$$y' = \sum_{m=0}^{\infty} m a_m x^{m-1}, \quad y'' = \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2}$$

$$0 = (1-x^2)y'' - 2xy' + n(n+1)y$$

$$\Rightarrow (1-x^2) \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} - 2x \sum_{m=0}^{\infty} m a_m x^{m-1} + n(n+1) \sum_{m=0}^{\infty} a_m x^m = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} + \sum_{m=0}^{\infty} \left[-m(m-1) - 2m + n(n+1) \right] a_m x^m = 0$$

Coefficient of x^s

$$(s+2)(s+1) a_{s+2} + (-s(s-1) - 2s + n(n+1)) a_s = 0$$

$$a_{s+2} = \frac{s(s-1) + 2s - n(n+1)}{(s+2)(s+1)} a_s$$

$$\begin{aligned} s(s-1) + 2s - n(n+1) &= s^2 - s + 2s - n^2 - n \\ &= s^2 - ns - n^2 + ns - n + s \\ &= s(s-n) + n(s-n) + (s-n) \\ &= (s+n)(s-n) + (s-n) \\ &= (s-n)(s+n+1) \end{aligned}$$

$$a_{s+2} = \frac{-(n-s)(s+n+1)}{(s+2)(s+1)} a_s$$

$$\frac{n(n+2)}{(n+2)(n+1)}$$

$$s=0 \quad a_2 = \frac{-n(n+1)}{2 \times 1} a_0$$

$$s=2 \quad a_4 = \frac{-(n-2)(n+3)}{4 \times 3} a_2$$

$$a_4 = \frac{n(n-2)(n+1)(n+3)}{4 \times 3 \times 2 \times 1} a_0$$

$$s=4 \quad a_6 = \frac{-n(n-2)(n-4)(n+1)(n+3)(n+5)}{6!} a_0$$

$$s=1 \quad a_3 = \frac{-(n-1)(n+2)}{3 \times 2} a_1$$

$$s=3 \quad a_5 = \frac{-(n-3)(n+4)}{5 \times 4} a_3$$

$$a_5 = \frac{(n-3)(n-1)(n+2)(n+4)}{5 \times 4 \times 3 \times 2} a_1$$

$$s=5 \quad a_7 = \frac{-(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!} a_1$$

$$y = \sum_{m=0}^{\infty} a_m x^m = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + a_7 x^7 + \dots$$

$$y = a_0 \left(1 - \frac{n(n+1)}{2!} x^2 + \frac{n(n-2)(n+1)(n+3)}{4!} x^4 - \frac{n(n-2)(n-4)(n+1)(n+3)(n+5)}{6!} x^6 + \dots \right)$$

$$+ a_1 \left(x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} x^5 - \frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!} x^7 + \dots \right)$$

$$y = a_0 y_1(x) + a_1 y_2(x)$$

* $y_1(x)$ and $y_2(x)$ are linearly independent.
Legendre's functions.

• If n is even, then $y_1(x)$ is a polynomial of degree n

$$P_n(x) = a_0 + a_2 x^2 + a_4 x^4 + \dots + a_n x^n$$

• If n is odd, then $y_2(x)$ is a polynomial of degree n

$$P_n(x) = a_1 x + a_3 x^3 + \dots + a_n x^n$$

These polynomials are called Legendre's polynomials.

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$$P_n(x) \begin{cases} \nearrow n \text{ is even} & a_0 + a_2 x^2 + a_4 x^4 + \dots + a_n x^n \\ \searrow n \text{ is odd} & a_1 x + a_3 x^3 + a_5 x^5 + \dots + a_n x^n \end{cases}$$

$$\circ \quad a_{s+2} = \frac{-(n-s)(n+s+1)}{(s+2)(s+1)} a_s$$

$$* \quad a_s = \frac{-(s+2)(s+1)}{(n-s)(n+s+1)} a_{s+2}$$

$$a_0 = \frac{-2 \times 1}{2 \times 3} a_2$$

$$a_0 = \frac{-2}{6} \times \frac{3}{2} = -\frac{1}{2}$$

$$* \quad a_n = \frac{(2n)!}{2^n \cdot (n!)^2}$$

$$P_0(x) = a_0 = \frac{1!}{2^0 \cdot (0!)^2} = 1 \quad \boxed{P_0(x) = 1}$$

$$P_1(x) = a_1 x = x \quad \boxed{P_1(x) = x}$$

$$a_1 = \frac{2!}{2^2 \cdot (1!)^2} = 1$$

$$P_2(x) = a_0 + a_2 x^2 = \frac{1}{2} + \frac{3}{2} x^2$$

$$a_2 = \frac{4!}{2^2 \cdot (2!)^2} = \frac{1 \times 2 \times 3 \times 4}{4 \times 2 \times 2}$$

$$P_3(x) = a_1 x + a_3 x^3 = \frac{3}{2} x + \frac{5}{2} x^3$$

$$\left\{ P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{3}{2} x^2 - \frac{1}{2}, \quad P_3(x) = \frac{5}{2} x^3 - \frac{3}{2} x \right\}$$

$$\textcircled{1} \quad \int_{-1}^1 P_0(x) P_1(x) dx = 0$$

$$\textcircled{4} \quad \int_{-1}^1 P_1(x) P_2(x) dx = \int_{-1}^1 \left(\frac{3}{2} x^3 - \frac{1}{2} x \right) dx = 0$$

$$\textcircled{2} \quad \int_{-1}^1 P_0(x) P_2(x) dx = \left[\frac{3}{2} \cdot \frac{x^3}{3} - \frac{1}{2} x \right]_{-1}^1 = 0$$

$$\textcircled{5} \quad \int_{-1}^1 P_2(x) P_3(x) dx = 0$$

$$\textcircled{3} \quad \int_{-1}^1 P_1(x) P_3(x) dx = \left[\frac{5}{2} \cdot \frac{x^4}{4} - \frac{3}{2} \cdot \frac{x^2}{2} \right]_{-1}^1 = 0$$

$$\textcircled{6} \quad \int_{-1}^1 P_1(x) P_3(x) dx = 0$$

$$\textcircled{3} \int_{-1}^1 P_0(x) P_3(x) dx = \left. \frac{5}{2} \cdot \frac{x^4}{4} - \frac{3}{2} \cdot \frac{x^2}{2} \right|_{-1}^1 = 0 \quad \textcircled{6} \int_{-1}^1 P_1(x) P_3(x) dx = 0$$

$$* \int_{-1}^1 P_n(x) P_m(x) dx = 0$$

• $P_n(x) \rightarrow$ Inner product space of polynomials of degree $\leq n$ with $\int_{-1}^1 p(x) q(x) dx = \langle p(x), q(x) \rangle$

• $P_3(x) \rightarrow \{ \overset{1}{P_0(x)}, \overset{1}{P_1(x)}, P_2(x), P_3(x) \} \rightarrow$ orthogonal Basis

• Legendre polynomials $\{ P_0(x), P_1(x), \dots, P_n(x) \}$ forms an orthonormal basis of inner product space of polynomial of degree $\leq n$ with inner product $\langle p(x), q(x) \rangle = \int_{-1}^1 p(x) q(x) dx$.

question: Is it orthonormal basis too? $\rightarrow$ (No)

$$\langle P_0(x), P_0(x) \rangle = \int_{-1}^1 P_0(x) P_0(x) dx = \int_{-1}^1 1 \cdot dx = x \Big|_{-1}^1 = 1 - (-1) = 2.$$

$$\langle P_n(x), P_n(x) \rangle = \int_{-1}^1 P_n^2(x) dx = \frac{2}{2n+1}$$

• second order ODE

$$y'' + p(x) y' + q(x) y = 0 \quad \textcircled{1}$$

• $x = x_0$ is ordinary point if $\lim_{x \rightarrow x_0} p(x)$, $\lim_{x \rightarrow x_0} q(x)$ both exists and are finite

* Eqn (1) has power series solution.

* Eqn ① has power series solution.

• If x_0 is Not ordinary point, then x_0 is a singular point.

* Regular singular point :- $x = x_0$ is called regular singular point if

$\lim_{x \rightarrow x_0} (x - x_0) p(x)$ and $\lim_{x \rightarrow x_0} (x - x_0)^2 q(x)$ exists and is finite.

Example :-

$x^2 y'' + x p(x) y' + Q(x) y = 0$, where $p(x)$ and $Q(x)$ are continuous functions. ①

$\lim_{x \rightarrow 0} \frac{p(x)}{x} \rightarrow \infty$, $\lim_{x \rightarrow 0} \frac{Q(x)}{x^2} \rightarrow \infty \Rightarrow x=0$ is Not ordinary point.

$\lim_{x \rightarrow 0} x \cdot \frac{p(x)}{x}$ exists & $\lim_{x \rightarrow 0} x^2 \cdot \frac{Q(x)}{x^2}$ exists $\Rightarrow x=0$ is a regular singular point.

$\rightarrow$ Suppose $y(x) = x^x \sum_{k=0}^{\infty} a_k x^k$ is solution of ①.

$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$ is solution of ①

$$y'(x) = \sum_{k=0}^{\infty} (k+r) a_k x^{k+r-1}$$

$$y''(x) = \sum_{k=0}^{\infty} (k+r)(k+r-1) a_k x^{k+r-2}$$

Then constant term = $\boxed{r(r-1) + p_0 r + q_0}$ $\rightarrow$ indicial Equation

where $p_0 = \lim_{x \rightarrow 0} x p(x)$

$q_0 = \lim_{x \rightarrow 0} x^2 Q(x)$

quadratic Eqn in r .

1, 1, 1 roots r_1 and r_2

Egn in r

Case I :- Indicial Egn has distinct roots r_1 and r_2
 Δ $r_1 - r_2$ is not an integer

then

$$y_1(x) = x^{r_1} \sum_{k=0}^{\infty} a_k x^k \quad \text{and} \quad y_2 = x^{r_2} \sum_{k=0}^{\infty} b_k x^k$$

are two L.I. solutions.

Case II :- Indicial Egn has same roots r_1, r_1

$$y_1(x) = x^{r_1} \sum_{k=0}^{\infty} a_k x^k$$

$$y_2(x) = y_1(x) \ln(x) + x^{r_1} \sum_{k=1}^{\infty} b_k x^k$$

Case III :- Distinct roots r_1 and r_2 , $r_1 - r_2$ is an integer. $\rightarrow$ $r_1 > r_2$

$$y_1(x) = x^{r_1} \sum_{k=0}^{\infty} a_k x^k$$

$$y_2(x) = y_1(x) \ln(x) + x^{r_2} \sum_{k=0}^{\infty} b_k x^k$$

